WORKED SOLUTION

STEP III 1989, Question 8

Pure • Coupled Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q8

The key idea

The first equation gives y in terms of x and $\mathrm{d}x/\mathrm{d}t$. So if we differentiate it once more, the $\mathrm{d}y/\mathrm{d}t$ that appears can be replaced using the second equation, and y drops out entirely. That leaves a second-order linear equation with constant coefficients. Its auxiliary equation has a repeated root, and one of the forcing terms matches that root. This resonance is the real test in the question.

Solution

Eliminating y

Write dots for derivatives with respect to t. Differentiating $\dot{x} = 4(x - y)$ gives

$$\ddot{x} = 4\dot{x} - 4\dot{y} = 4\dot{x} - 4x + 48\left(\mathrm{e}^{2t} + \mathrm{e}^{-2t}\right),$$

using the second equation for $\dot{y}$. Hence

$$\boxed{\ddot{x} - 4\dot{x} + 4x = 48\left(\mathrm{e}^{2t} + \mathrm{e}^{-2t}\right),}$$

which contains no y.

Solving for x

Complementary function. The auxiliary equation $m^2 - 4m + 4 = (m-2)^2 = 0$ has the repeated root $m = 2$, so the complementary function is

$$x_\mathrm{c} = (A + Bt)\,\mathrm{e}^{2t}.$$

Particular integral. Since the equation is linear, we can deal with the two forcing terms one at a time.

For $48\mathrm{e}^{-2t}$, try $x = D\mathrm{e}^{-2t}$. Substituting gives $D(4 + 8 + 4)\mathrm{e}^{-2t} = 48\mathrm{e}^{-2t}$, so $D = 3$.

Ask yourself Why won't Ce^{2t}, or even Cte^{2t}, work for the $48e^{2t}$ term?

Both e^{2t} and te^{2t} are already in the complementary function, so the left-hand side sends them to 0. Because 2 is a *double* root, we have to go up to t^2e^{2t}. Try $x = Ct^2e^{2t}$. A neat way to substitute is to write $x = ue^{2t}$. Then

$$\dot{x} = (\dot{u} + 2u)e^{2t}, \qquad \ddot{x} = (\ddot{u} + 4\dot{u} + 4u)e^{2t},$$

so

$$\ddot{x} - 4\dot{x} + 4x = \ddot{u}\,e^{2t}.$$

With $u = Ct^2$ this is $2Ce^{2t}$, so $2C = 48$ and $C = 24$. (The substitution also shows why the complementary function is $(A + Bt)e^{2t}$: we need $\ddot{u} = 0$.)

So the general solution is

$$x = (A + Bt)\,e^{2t} + 24t^2e^{2t} + 3e^{-2t}.$$

Initial conditions. We are given only $x(0) = 0$ and $y(0) = 0$. The second-order equation needs $\dot{x}(0)$, and the first original equation supplies it: $\dot{x}(0) = 4(x(0) - y(0)) = 0$.

From $x(0) = 0$ we get $A + 3 = 0$, so $A = -3$. Differentiating,

$$\dot{x} = (B + 2A + 2Bt + 48t + 48t^2)e^{2t} - 6e^{-2t},$$

so $\dot{x}(0) = B + 2A - 6 = 0$, giving $B = 12$. Therefore

$$\boxed{x = \left(24t^2 + 12t - 3\right)e^{2t} + 3e^{-2t}.}$$

Finding y

There is no need to solve another differential equation, because the first equation gives $y = x - \frac{1}{4}\dot{x}$. With $A = -3$ and $B = 12$,

$$\dot{x} = \left(48t^2 + 72t + 6\right)e^{2t} - 6e^{-2t},$$

so

$$\begin{aligned} y &= \left(24t^2 + 12t - 3\right)e^{2t} + 3e^{-2t} - \left(12t^2 + 18t + \tfrac{3}{2}\right)e^{2t} + \tfrac{3}{2}e^{-2t} \\ &= \left(12t^2 - 6t - \tfrac{9}{2}\right)e^{2t} + \tfrac{9}{2}e^{-2t}. \end{aligned}$$

That is,

$$\boxed{y = \tfrac{3}{2}\left(8t^2 - 4t - 3\right)e^{2t} + \tfrac{9}{2}e^{-2t}.}$$

Checks

- At $t = 0$: $y(0) = -\frac{9}{2} + \frac{9}{2} = 0$. This was never imposed directly, so it is a genuine check.
- The second equation, which we used only to eliminate y, should hold too. Differentiating gives
$$\dot{y} = \left(24t^2 + 12t - 15\right)\mathrm{e}^{2t} - 9\mathrm{e}^{-2t},$$
and
$$x - 12\left(\mathrm{e}^{2t} + \mathrm{e}^{-2t}\right) = \left(24t^2 + 12t - 15\right)\mathrm{e}^{2t} - 9\mathrm{e}^{-2t}.$$
These agree.
- Numerically, stepping the original pair forward from $(0, 0)$ with a Runge–Kutta method gives $x(1) \approx 244.2449$ and $y(1) \approx 11.6926$. The formulae give $33\mathrm{e}^2 + 3\mathrm{e}^{-2}$ and $\frac{3}{2}\mathrm{e}^2 + \frac{9}{2}\mathrm{e}^{-2}$, which have the same values.

Common slips

- Trying $Ct\,\mathrm{e}^{2t}$ for the resonant term. The root 2 is repeated, so we need $t^2\mathrm{e}^{2t}$.
- Not seeing where the second initial condition comes from. The value $\dot{x}(0) = 0$ is hidden in the first equation together with $x(0) = y(0)$.
- Solving a separate second-order equation for y. That brings in two new constants, which then have to be matched to x. Using $y = x - \frac{1}{4}\dot{x}$ avoids this.

Beyond STEP

Eliminating y is one way to solve a linear system. The undergraduate approach writes it as $\dot{\mathbf{v}} = M\mathbf{v} + \mathbf{f}(t)$ with $M = \begin{pmatrix} 4 & -4 \\ 1 & 0 \end{pmatrix}$. Here M has the repeated eigenvalue 2 but only one eigenvector, so it cannot be diagonalised, and that is exactly where the $t\,\mathrm{e}^{2t}$ terms come from. The Jordan normal form and the matrix exponential e^{Mt} make this precise. Boyce and DiPrima, *Elementary Differential Equations*, covers systems of linear equations, including the repeated-eigenvalue case.

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